MATE664 L17

Coarsening & rippening phenomena

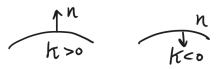
Coarsening: ① Particle size: $\begin{cases} \text{smaller shrink} \\ \text{larger grow} \end{cases}$ (Conserved system)

② particle size distribution e.g. density functional

$\rho(r)$ has self-similarity

Explanation: capillarity (surface-energy related phenomena)

Definition of surface norm vector & curvature



But $\kappa \cdot \vec{n}$ is same value

→ Low free energy / chem pot. / pressure

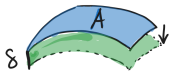
→ High free energy / .... / ....

Effective curvature H or K is average over two principle curvatures

$H = \frac{1}{2}(\kappa_1 + \kappa_2)$

What if one κ_1 & κ_2 is negative? ⇒ saddle point in transition state theory

Driving force from capillarity? $\gamma(\kappa_1 + \kappa_2)$ For a sphere, we have:



$$\Delta F = \frac{\Delta E}{\Delta V} = \frac{\gamma \Delta A}{\Delta V} = \frac{\gamma \Delta(4\pi R^2)}{\Delta(\frac{4}{3}\pi R^3)}$$

$$= \frac{2 \cdot \gamma \cdot 4\pi R}{4\pi R^2} = 2\gamma \frac{1}{R} \Rightarrow \gamma \cdot \frac{\cancel{2}}{\cancel{2}}(\kappa_1 + \kappa_2)$$

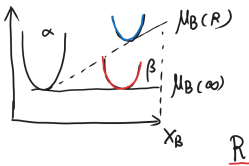
Generalized

The LSW model explained:

Key component: curvature ⇒ higher interfacial chemical potential

Due to capillarity, additional free energy $\frac{2\gamma\Omega_B}{R}$ is added for each atom. Why $C^{eq}(R)$ higher than $C^{eq}(\infty)$?

Gibbs-Thompson effect



We have

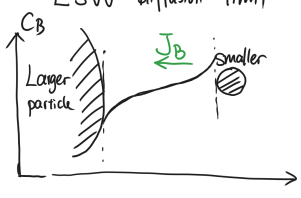
$$\mu = \mu^0 + k_B T \ln\left(\frac{C}{C_0}\right)$$

assuming ideal

R smaller ⇒ higher μ_B ⇒ higher C_B^{eq}

(thought experiment: if curvature is large enough ⇒ $R \approx 0$
there will still to have B molecules around the surface
$C_B(R \to 0)$ will be approaching pure material limit)

LSW diffusion limit



mass transfer happens from smaller → larger
(remember higher $\mu_B \xrightarrow{J_B}$ lower μ_B)

But in the LSW picture it is usually distinguished wether diffusion is fast enough compared with growth

LSW theory deals with a conserved material field, so $C_{eq}(\infty)$ depends on current average size $\langle R \rangle$

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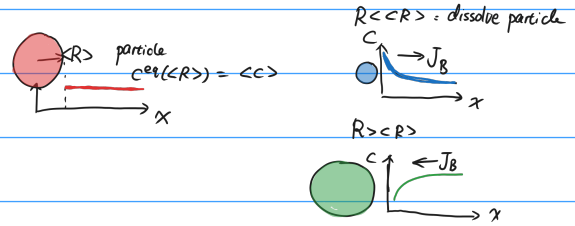
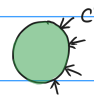


In diffusion-limited regime, the concentration field has accumulation & depletion regions

Mass balance around a spherical particle



$$J_B \cdot A = -\frac{dR}{dt} \cdot A \cdot \Omega_B$$

Solution for J_B in spherical coordinate $\Rightarrow \frac{C_{eq} - \langle C \rangle}{R} \tilde{D}$ (Volume $\cdot \Omega_B$ = number concentration)

$$\hookrightarrow \frac{dR}{dt} = -\tilde{D} \frac{c^{eq}(R) - \langle c \rangle}{R} \Omega_B$$

- Substitute $c^{eq}(R) = C_\infty^{eq} \exp\left(\frac{2\gamma\Omega_B}{k_B T}\right) \simeq C_\infty^{eq}\left(1 + \frac{2\gamma\Omega_B}{k_B T}\right)$
- Conservation $\sum R^2 \frac{dR}{dt} = 0$ (total volume conserved)

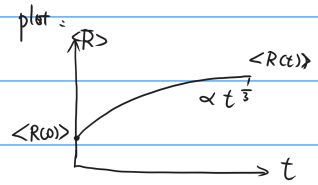
$$\hookrightarrow \text{Power law} \quad \frac{dR}{dt} \propto \frac{2\tilde{D}\gamma\Omega_B^2 C^{eq}(\infty)}{k_B T} \frac{1}{R}\left(\frac{1}{\langle R \rangle} - \frac{1}{R}\right)$$

Because $\frac{dR}{dt} \propto \frac{1}{R^2}$, can expect $R^3(t) \propto t$ powerlaw

Actual result

$$\underbrace{\langle R(t) \rangle}_{\text{average size at } t}{}^3 - \underbrace{\langle R(0) \rangle}_{\text{average size at } t=0}{}^3 = \underbrace{\frac{8\tilde{D}\gamma\Omega_B^2 C^{eq}(\infty)}{9 k_B T}}_{K_D \text{ (factor)}} t$$

Usual plot:



Another regime: source limited $\frac{dR}{dt} = k[c(R) - c^{eq}(R)]\Omega_B$

$\Uparrow$ mass transfer coeff, R-independent

To get final form for source-limited regime

$$\langle R^2(t) \rangle - \langle R^2(0) \rangle = \underbrace{\frac{64 k c^{eq}(\infty)\gamma\Omega^2}{81 k_B T}}_{K_S} t$$

Comparison:

$$K_D = \frac{8\tilde{D}\gamma\Omega_B^2 C^{eq}(\infty)}{k_B T}$$

$$K_S = \frac{64 k c^{eq}(\infty)\gamma\Omega^2}{81 k_B T}$$

Difference

$\tilde{D}$ unit m^2/s

k unit m/s

$\Downarrow$ Different power law

motivation: Can we have other $R \propto t^p$ power law?